

Supplementary Material

Transparent Public-Data Prioritization for Medicaid Long-Term-Care Oversight: Evidence From Two States

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This document contains Supplementary Appendices A through H for the manuscript “Transparent Public-Data Prioritization for Medicaid Long-Term-Care Oversight: Evidence From Two States.” All section references (e.g., “Section 3.4”) refer to the main manuscript. All appendices are cited in the text of the main manuscript.

Appendix A: Oregon’s Facility-Type Confounding Diagnosis

Section 3.4 reports that Oregon’s shared-telephone clustering threshold required facility-type-aware recalibration. This appendix reports the full diagnostic evidence behind that conclusion: a categorization of all 49 raw shared-phone clusters identified in the Oregon analysis by whether the paired facilities shared a physical address and by facility type.

Tier	Definition	Count	Interpretation
A: Campus Pairs	Same physical address, non-AFH facility types	25	Co-located license splits on a single campus (e.g., a licensed assisted-living wing paired with a separately licensed memory-care wing at the same street address). Not indicative of multi-site operator concentration; a structural artifact of how Oregon licenses distinct care levels within one physical development.
B: Brand Pairs/Clusters	Different addresses, non-AFH facility types, shared operating-brand naming	13	Likely reflects a multi-building senior-living campus or a single corporate brand operating adjoining or nearby

Tier	Definition	Count	Interpretation
			developments. Ambiguous without ownership records. Includes the single five-license cluster responsible for every RF-08 flag.
C: AFH Operator Pairs	Different addresses, Adult Foster Home type, same individual operator of record on both licenses	9	The closest Oregon analogue to the construct the framework was built to detect: an individual licensed to operate more than one home under their own name. The fairest like-for-like comparison to the Washington pilot population.
D: Unexplained AFH Pairs	Different addresses, Adult Foster Home type, different named operators, sharing one telephone number	2	The most analytically interesting residual category: two separately named AFH operators, at two different addresses, sharing one administrative telephone number, with no address- or name-based explanation apparent from public data alone.

Of the 49 raw clusters, 11 (Tiers C and D) resemble the intended multi-site-operator construct the framework targets. The remaining 38 (Tiers A and B), including the single five-license cluster that produces every RF-08 flag, appear more consistent with Oregon's facility-licensing and branding structure than with an undisclosed operator-concentration signal. This is the specific, falsifiable evidence behind the recalibration claim in Section 3.4: not an assertion that the threshold needed adjustment, but a categorized accounting of why.

The same recalibration question was also tested across a documented threshold range, rather than searched for a value that reproduced a targeted result:

Threshold (shared licenses)	All facility types (n = 792)	AFH-only (n = 629)
2 or more	101 facilities (12.8%)	22 facilities (3.5%), 11 distinct clusters
3 or more (RF-08, as calibrated on Washington)	5 facilities (0.6%)	0 facilities
4 or more	5 facilities (0.6%)	0 facilities
5 or more	5 facilities (0.6%)	0 facilities

From threshold three upward, the all-facility-type curve is carried entirely by the single five-license Tier B cluster. The AFH-only curve never reaches the three-license threshold in this two-county dataset. Restricting the comparison to the 629-facility Adult Foster Home subset, the facility type matching Washington's pilot population, identifies a pair-level sensitivity pattern at the two-license threshold, which Section 3.4 characterizes as a sensitivity result, not a validated threshold.

Exploratory comparison: enforcement/investigation prevalence, clustered vs. nonclustered facilities

This is an exploratory, hypothesis-generating comparison, not a predictive-validity test — predictive validity remains untested for every indicator in this framework (Section 2.6). It asks a narrower question than the confounding diagnosis above: among facilities not restricted to the RF-08 threshold, does public enforcement or investigation prevalence differ between facilities in *any* shared-contact cluster and facilities in none, and does that comparison depend on the facility-type or county composition of who happens to be clustered? Computed via `scripts/build_cluster_enforcement_exploratory.py`: 2×2 tables (clustered vs. not) × (≥1 enforcement action / ≥1 investigation), crude and Mantel-Haenszel-adjusted (stratified by County for Washington, by facility type for Oregon), each run twice — on all facilities and again excluding facilities with no observable public record (`n_docs_total == 0`) — and reported with absolute prevalence, risk ratio, and odds ratio, each with a 95% CI, regardless of direction.

State	Outcome	Facilities excl. no- record?	n clustered / % with event	n nonclustered / % with event	Crude RR (95% CI)	Facility- type/ county- adjusted RR (95% CI)
Washington	Enforcement	No	164 / 6.7%	3,293 / 4.7%	1.43 (0.79–2.59)	1.43 (0.79–2.60)
Washington	Enforcement	Yes	118 / 9.3%	2,052 / 7.5%	1.24 (0.69–2.23)	1.23 (0.68–2.20)
Washington	Investigation	No	164 / 17.1%	3,293 / 16.9%	1.01 (0.71–1.42)	1.01 (0.71–1.42)
Washington	Investigation	Yes	118 / 23.7%	2,052 / 27.2%	0.87 (0.63–1.22)	0.86 (0.62–1.20)
Oregon	Enforcement	No	101 /	691 /	2.10 (1.53–	0.66 (0.48–

State	Outcome	Facilities excl. no- record?	n clustered / % with event	n nonclustered / % with event	Crude RR (95% CI)	Facility- type/ county- adjusted RR (95% CI)
	nt		34.7%	16.5%	2.88)	0.92)
Oregon	Enforceme nt	Yes	99 / 35.4%	575 / 19.8%	1.78 (1.30– 2.44)	0.64 (0.46– 0.89)
Oregon	Investigati on	No	101 / 54.5%	691 / 24.0%	2.27 (1.82– 2.83)	1.16 (0.91– 1.49)
Oregon	Investigati on	Yes	99 / 55.6%	575 / 28.9%	1.92 (1.55– 2.39)	1.11 (0.87– 1.42)

Full stratum-level 2×2 tables, odds ratios, and both record-filter conditions are in `anonymized_data/cluster_enforcement_exploratory_detailed.csv` and `_summary.csv`.

Washington’s comparison is essentially unchanged by county adjustment — crude and adjusted estimates are nearly identical for both outcomes, all confidence intervals crossing 1. Oregon’s is not: the crude enforcement comparison shows clustered facilities more than twice as likely to carry an enforcement action (RR 2.10, 95% CI 1.53–2.88) — but stratified by facility type, this reverses to a statistically significant *lower* prevalence among clustered facilities (RR 0.66, 95% CI 0.48–0.92). The mechanism is directly visible in the stratum-level data: within every one of Oregon’s four facility types individually, clustered facilities show lower or comparable enforcement prevalence than nonclustered facilities of the same type (0 of 22 vs. 66 of 607 for Adult Foster Homes; 13 of 33 vs. 17 of 27 for Assisted Living; 1 of 3 vs. 8 of 19 for Nursing Facilities; 21 of 43 vs. 23 of 38 for Residential Care Facilities) — but because clustered facilities are disproportionately concentrated in Oregon’s higher-baseline-enforcement facility types (Assisted Living, Residential Care) rather than its lower-baseline-enforcement type (Adult Foster Home), the pooled crude comparison mechanically inherits that composition difference and points the opposite direction from every within-type comparison.

This is not evidence that shared-contact clustering identifies “safer” facilities, and this appendix does not make that claim — the direction, magnitude, and even statistical significance of this comparison changes depending on whether facility-type composition is accounted for, which is itself the finding: an unadjusted association between a structural indicator and an oversight outcome can be an artifact of population composition rather than a property of the indicator, precisely the population-validity distinction Section 2.6 defines and Section 3.4 documents for the clustering threshold itself. No claim in the main manuscript relies on this comparison; it is reported here as a further, independent illustration of the same lesson, not a new result requiring a change to any indicator’s disposition.

Appendix B: Full RF-01 Through RF-20 Evidentiary Classification, Both States

Sections 3.3-3.4 report the two-state facility-level and case-level implementation results; the two-state assessment separately produced heterogeneous outcomes across the 20 indicators: some directly operationalized at facility level, others documented by a case or a system-level audit, others only partially operationalized, contextually assessed, publicly testable but not completed, or requiring internal data no public-data method can reach. This appendix reports the full indicator-by-indicator classification behind that accounting: Washington’s original five-way testability label and Oregon’s eight-tier evidentiary status, side by side, plus a Combined Evidentiary Classification column.

The Washington and Oregon status labels describe whether public data could in principle support an indicator, not whether this study actually computed or assessed it; a label of “Testable” or “Partially Testable” can describe an indicator that was fully computed, only sampled, or not computed at all. The Combined Evidentiary Classification is therefore not derived from those labels directly. Each row was instead audited against the underlying status notes describing what was actually done in each state (WA_RedFlag_Testability_Matrix_v2.csv’s Status_Notes column and OR_RF_Evidence_Matrix.csv’s Basis column), assigned to one of seven canonical evidentiary categories per state, then mapped to the most directly evidenced category available across the two states under a documented rule in `scripts/build_combined_matrix.py`, which records the audit rationale for every row. This table answers one narrow question — what was operationalized, partially operationalized, documented at case/system level, or inaccessible in Washington and Oregon — and is synchronized here with the final analytic architecture (Appendix G) so the two do not drift apart; it does not duplicate Appendix G/H’s source citations or corroboration evidence, which belong there. A **Post-Audit Status** column carries that reconciliation: whether the row’s original operationalization classification still applies unchanged, or the row is now retired, split, or reframed as a proxy.

RF	Indicator	Washington Status	Oregon Status	Combined Classification	Post-Audit Status
RF-01	Rapid provider/facility growth	Testable (program/sector level)	Publicly Testable, Not Completed	Publicly Testable, Not Completed	Unchanged
RF-02	Spending-growth-disconnected from beneficiary-growth	Testable (program level)	Contextually Assessed	Partially Operationalized	Retired (Appendix G) — classification below is historical, not current
RF-03	Multi-entity/multi-facility operator	Partially Testable	Partially Operationalized	Partially Operationalized	Unchanged

RF	Indicator	Washington Status	Oregon Status	Combined Classification	Post-Audit Status
	concentration				
RF-04	Geographic concentration or out-of-state operator	Partially Testable	Publicly Testable, Not Completed	Publicly Testable, Not Completed	Recommend ed for retirement/redefinition (Appendix G)
RF-05	Incomplete administrative or enrollment data	Not Publicly Testable	Partially Operationalized	Partially Operationalized	Unchanged
RF-06	Excluded party involvement or linkage gap	Partially Testable	Publicly Testable, Not Completed	Partially Operationalized	Unchanged (operationalization); externally Strong (Appendix H)
RF-07	Concealed or opaque ownership / related-party control	Partially Testable	Publicly Testable, Not Completed	Publicly Testable, Not Completed	Unchanged
RF-08	Shared address, phone, or network (cluster threshold)	Testable (proxy)	Operationalized; recalibration required	Directly Operationalized	Non-additive proxy under RF-07 (Appendix G) — operationalization below reflects the measurement, not an independent construct
RF-09	Phantom, mailbox, or questionable address	Partially Testable	Contextually Assessed	Contextually Assessed	Unchanged
RF-10	Impossible hours or overlapping services	Not Publicly Testable	Internal Data Required	Internal Data Required	Unchanged

RF	Indicator	Washington Status	Oregon Status	Combined Classification	Post-Audit Status
RF-11	Maximum-unit or implausible service-volume billing	Not Publicly Testable	Internal Data Required	Internal Data Required	Unchanged
RF-12	Out-of-state or impossible-location billing	Not Publicly Testable	Case Supported	Case-Level Evidence	Recommended for retirement/redefinition (Appendix G) — the Oregon case below is related but does not match the final geotemporal mechanism
RF-13	Billing after death, absence, discharge, or ineligibility (renamed: billing not rendered or not eligible)	Not Publicly Testable	Internal Data Required	Internal Data Required	Unchanged classification; externally Strong (Appendix H)
RF-14A	Missing or insufficient supporting documentation (split from RF-14)	Not Publicly Testable	Case Supported	Case-Level Evidence	Split from RF-14 (Appendix G) — this row covers the missing-documentation half only
RF-14B	Cloned or fabricated documentation (split from RF-14)	Not Publicly Testable	Case Supported	Case-Level Evidence	Split from RF-14 (Appendix G) — this row covers the fabrication

RF	Indicator	Washington Status	Oregon Status	Combined Classification	Post-Audit Status
					half only, assessed independently of RF-14A
RF-15	Stolen credentials or unauthorized professional IDs	Not Publicly Testable	Not Established	Internal Data Required; Not Established	Unchanged
RF-16	Frontline referrals not acted upon or poorly triaged	Partially Testable	System Supported	System-Level Evidence	Unchanged (operationalization); externally Strong (Appendix H)
RF-17A	Prior enforcement, suspension, or sanction history (split from RF-17)	Partially Testable	Partially Operationalized	Directly Operationalized	Split from RF-17 (Appendix G) — background/descriptive, not independently scored for external convergence
RF-17B	Continued participation despite exclusion or sanction (split from RF-17)	Partially Testable	Partially Operationalized	Directly Operationalized	Split from RF-17 (Appendix G); externally Strong (Appendix H)
RF-18	Reactive oversight and insufficient capacity	Fully Testable	System Supported	System-Level Evidence	Unchanged
RF-19	Program design vulnerability	Testable (qualitative)	Contextually Assessed	Contextually Assessed	Unchanged
RF-20	Unresolved audit findings	Fully Testable	System Supported	System-Level Evidence	Unchanged classification;

RF	Indicator	Washington Status	Oregon Status	Combined Classification	Post-Audit Status
	or weak corrective-action follow-through				externally Strong (Appendix H)

RF-01 and RF-04 illustrate the distinction the audit draws: Washington’s licensing data contains a field that would support a facility-growth trend or an out-of-state-operator flag in principle, but neither trend was actually constructed in this study, so both are coded Publicly Testable, Not Completed rather than Directly Operationalized. RF-06 reflects an executed but inconclusive computation: Washington’s exclusion-list cross-match was actually run but could not reliably link facility and licensee names, not because the match was never attempted. RF-17A/B illustrate a stronger case still: Washington’s sanction and enforcement counts (165 facilities with an action, Section 3.3; 18 stop-placement orders, Appendix F) were in fact fully computed at facility level, so both rows are coded Directly Operationalized for what was measured in Washington/Oregon, independent of RF-17B’s separate external-convergence rating. RF-18 and RF-20 were strongly documented, but by state and federal audits rather than a facility-level computation, so both are coded System-Level Evidence rather than Directly Operationalized. RF-02’s row is retained here only to show what was operationalized before the derivation audit retired the construct; its Combined Classification is historical and should not be read as current status.

Appendix C: Oregon’s System-Level Oversight Evidence, in Full

Section 1.1 and 3.2 report Oregon’s SOQ evaluation headline figures (8% survey-timeliness compliance, 4,000+ case backlog). The full evaluation traces a four-stage chronology, reported here in full because the sequence itself, not only the headline numbers, is part of the evidentiary basis for classifying RF-16, RF-18, and RF-20 as System Supported.

Identified problem (February 2025). The first deliverable found the unit responsible for Assisted Living and Residential Care Facility licensing completed statutory two-year renewal surveys for only 8% of active facilities examined, and Nursing Facility survey compliance with its one-year window was 4.3%. It documented a complaint-investigation backlog exceeding 4,000 cases, with fewer than 200 closed in the month examined, and reliance on informal “Letters of Agreement” in place of formal licensing conditions Oregon statute requires, including for some findings of immediate jeopardy (Oregon Department of Human Services, 2025a).

Formal response (documented in the June 2025 follow-up, reviewing more than 220 legislative mandates, 30 Oregon Administrative Rules, and 115 staff surveys). Oregon’s documented response included revision of Immediate Jeopardy decision protocols, a stated intent to clarify Letters of Agreement policy, statutory-requirement training for staff, workload and staffing realignment, a new performance-measurement structure, and active work items within the Licensing Complaint Unit covering complaint triage and a suspicious-death procedure (Oregon Department of Human Services, 2025b).

Follow-up assessment milestone. The June 2025 deliverable is itself the milestone: a second, independently commissioned, more granular assessment that documented planned remediation and several initiated work items rather than letting the process stall after the initial diagnosis, with an explicit remediation estimate of roughly one year to rebuild the unit's core structures (Oregon Department of Human Services, 2025b).

Remaining outcome gap. No third deliverable documenting the rebuild's outcome had been published as of this manuscript's snapshot date, and the June 2025 report itself still found the licensing unit lacking a shared understanding of organizational responsibilities and not routinely meeting survey and complaint-investigation timelines (Oregon Department of Human Services, 2025b). The corrective-action plan is reported here as documented and adopted in principle, not as completed in practice.

Appendix D: Additional Washington System-Level Evidence

Beyond the HHS-OIG compliance audit reported in Section 3.3, Washington's own oversight infrastructure has been independently assessed multiple times, evidence this manuscript's main text does not have room to detail but which provides additional evidence of reactive or incomplete program-integrity controls from related perspectives.

The Washington State Auditor's Office found the same posture in two prior performance audits: its 2021 audit of the Health Care Authority's program-integrity function, and its 2023 audit of Medicaid managed-care program-integrity efforts, both describing oversight as reactive and complaint-driven rather than proactive and risk-based (Washington State Auditor's Office, 2021, 2023). A more recent state audit found more than \$5.6 million in inappropriate Medicaid payments and documented internal-control deficiencies related to provider eligibility and managed-care oversight (State of Washington, Office of Financial Management, 2026). A joint review by the Washington State Auditor's Office and the HHS Office of Inspector General separately identified deficiencies in concurrent Medicaid enrollment eligibility integrity (U.S. Department of Health and Human Services, Office of Inspector General, 2025). And, as of the original Washington pilot's writing, the State Auditor's Office had an active performance audit underway examining more than 600 DSHS vendors whose spending showed patterns of exponential growth between 2021 and 2025 (Washington State Auditor's Office, 2026); this active audit is cited only as evidence of ongoing oversight attention to vendor spending growth, since no findings had been issued as of this manuscript's snapshot date.

Separately, a March 2026 federal advisory documented a 330% increase in Bank Secrecy Act filings related to healthcare fraud between 2020 and 2025, flagging the use of shell companies and straw owners to conceal beneficial ownership, a concealment pattern relevant to RF-07's ownership-transparency construct (Financial Crimes Enforcement Network, 2026).

Appendix E: Statewide Enforcement Context, Both States

The county- and facility-level findings in this manuscript sit inside broader statewide enforcement records in both states, reported here for context; none of these statewide figures are attributable to either study county or facility type specifically.

Oregon's Medicaid Fraud Unit reports 348 criminal convictions and 156 civil settlements since 2010 across all Medicaid provider types, with more than \$146 million recovered (Oregon

Department of Justice, 2026). Oregon’s public response to the March 2026 congressional inquiry reported approximately \$160 million in improper Medicaid payments recovered from 2021 through 2025 (Governor Kotek, 2026). Minnesota, the framework’s derivation state, saw a Department of Justice takedown in May 2026 charging 15 defendants in a scheme exceeding \$90 million (U.S. Department of Justice, 2026), consistent with the broader documented audit and enforcement corpus Section 2.1 draws on.

Appendix F: Additional Washington Methodological Detail

Section 3.3 reports Washington’s headline structural counts. This appendix reports supporting methodological detail from the original Washington pilot that strengthens confidence in those counts but did not fit the main text.

Data completeness. Of the 3,457 pilot facilities, 1,287 (37.2%) had no public record of any kind, no inspection, relicensing action, complaint investigation, or enforcement action, within the study’s available public-record window. This is a data-visibility limit, not evidence that those facilities had no oversight history; it is reported as a documented gap, consistent with how this manuscript treats Oregon’s own missingness in Section 3.4.

Independent corroboration of shared-contact clusters. Of the 72 Washington shared-contact clusters, 70 (97.2%) had at least one additional administrative linkage beyond the shared telephone number: a shared DSHS administrative contact, a shared Secretary of State registered agent or principal-office address, a shared corporate governor, or a shared mailing address. Of those 70, only 26 clusters (14 via a shared Secretary of State registered agent, 12 via shared corporate governors), 36.1% of the 72 total clusters, were corroborated through a source independent of the licensing system itself; the remaining linkages drew on the same DSHS database the phone-based clustering came from, and so provide confirmatory but not independent support. This cross-check does not establish beneficial ownership; the shared-contact proxy was accompanied by at least one additional administrative linkage in most clusters, reducing but not eliminating the possibility of incidental number-sharing.

Enforcement severity and county-level detail. Of the 165 Washington facilities with at least one public enforcement action, 60 facilities—1.7% of the full 3,457-facility sample—had two or more enforcement actions, and 18 facilities (0.5% of the full sample) received a stop-placement order, the most severe action recorded. Within the three-county pilot, shared-contact cluster share ranged from 3.6% in Spokane County (22 of 610 facilities) to 5.3% in King County (95 of 1,795), with Pierce County at 4.5% (47 of 1,052). Pierce County separately showed the highest facility density per senior (approximately 7.2 per 1,000 residents 65 and older) alongside the highest enforcement (5.8%) and complaint (approximately 18%) rates of the three counties. This descriptive co-occurrence was not tested for statistical association and should not be interpreted as evidence that density caused greater enforcement activity. Combining cluster membership and enforcement-action flags, 318 facilities (9.2% of the pilot universe) were flagged by at least one of the two signals.

Appendix G: RF Specification Registry

This appendix reports the outcome of a four-stage evidentiary audit of the 20-indicator framework: a derivation audit tracing each indicator to an exact passage (or lack of one) in the

three documents spanning two Minnesota documentary lineages that generated the initial taxonomy (Section 2.1); a prespecified external-corroboration search against federal agencies and five comparison states (California, Michigan, Oregon, Texas, Washington); a corpus-wide adjudication assigning a final convergence rating and disposition to each indicator; and, for RF-08 specifically, an empirical corroboration study of Oregon’s shared-contact clusters. The audit was designed to be capable of contradicting, retiring, or dividing an indicator, not only confirming the original twenty, and it did all three: one indicator was retired because a primary source contradicted it, and two were split into subconstructs once the evidence showed the original wording bundled evidentially distinct claims.

Table G1 reports the final construct wording (where revised), the strength of its Minnesota derivation anchor, the outcome of the external-corroboration search, and the disposition that follows from combining the two. “Anchor status” and “external convergence” are independent axes—an indicator can have a strong original anchor and no external corroboration (most of this framework), weak original anchor evidence and strong external corroboration (RF-06, RF-17B), or fail both (RF-04, RF-12). Neither axis alone determines disposition.

RF	Final construct (as revised)	MN derivation anchor status	External convergence outcome	Disposition
RF-01	Rapid provider enrollment or facility growth	Directly stated (2009 OLA: 150%/174% unlicensed-agency growth)	None found (federal + 5-state complete; no source identified to backward-cite from)	Retain — anchor-derived; no external corroboration located
RF-02	<i>(retired)</i> Spending growth disconnected from beneficiary growth	Contradicted — 2009 OLA data shows growth tracked recipients, not diverging from them	Not applicable	Retired. Identifier permanently preserved, not reused
RF-03	Multi-entity/multi-facility operator concentration	Reasonable abstraction (2017 OLA: multiple provider IDs per ownership entity — related to, not identical to, multi-license concentration)	None found (federal + 5-state complete)	Retain — anchor-derived (abstraction-level)
RF-04	Geographic concentration or out-of-state	Not derivable from either anchor (a DOJ	None found (federal + 5-state complete; no	Recommended for retirement or narrower

RF	Final construct (as revised)	MN derivation anchor status	External convergence outcome	Disposition
	operator	2026 passage on Pennsylvania residents committing HSS fraud in Minnesota, considered and rejected: it documents the geography of individual participants, not a facility operated or controlled from out of state)	source identified to backward-cite from)	redefinition — see false-positive discussion below
RF-05	Incomplete administrative or enrollment data	Directly stated (2017 OLA — cleanest anchor in the set)	Strong — HHS-OIG, full text verified: most of 39 states did not provide complete or accurate T-MSIS managed-care payment data, two provided none (OEI-02-19-00180, 2021, p. 1)	Retain — Strong
RF-06	Excluded party involvement or linkage gap	Weak/indirect (2009 OLA documents the screening step, not evasion of it)	Strong — two independent national HHS-OIG datasets (2015, 2020)	Retain — Strong; external evidence substantially exceeds the derivation anchor
RF-07	Concealed or opaque ownership / related-party control	Directly stated (DOJ EIDBI case — undisclosed ownership of two autism	Limited (a Medicare-setting OIG finding and an academic paper, neither an exact-setting	Retain — Limited

RF	Final construct (as revised)	MN derivation anchor status	External convergence outcome	Disposition
		centers)	Tier 1 source)	
RF-08	Shared address, phone, registered agent, or network (WA/OR public-data proxy for RF-07)	Framed as RF-07's operational proxy, not independently MN-derived	Empirically tested directly (Oregon, 49 clusters; see below)	Retain as an operational proxy — not scored as an independent construct
RF-09	Phantom, mailbox, virtual, or questionable address	Not derivable from either anchor	Limited/related (exact-mechanism GAO finding, but Medicare/PECO S setting, and blocked from full-text verification)	Retain provisionally — Limited/related
RF-10	Impossible hours or overlapping services (>24 hrs/day)	Directly stated (2009 OLA: 314 instances, 152 consecutive, \$308,000)	None qualifying found	Retain — anchor-derived; no external corroboration located
RF-11	Maximum-unit or implausible service-volume billing	Reasonable abstraction, weak (2009 OLA: statutory cap only)	None found in a Medicaid setting (Medicare-only parallels)	Retain — anchor-derived (weak abstraction)
RF-12	Out-of-state or impossible-location billing	Not derivable from either anchor (a 2009 OLA passage on billing during beneficiary hospitalization, considered and rejected: it documents an incompatible-service-setting/beneficiary-unavailability	None found (federal + 5-state complete; no source identified to backward-cite from)	Recommended for retirement, or retention only as a narrowly redefined research hypothesis

RF	Final construct (as revised)	MN derivation anchor status	External convergence outcome	Disposition
		mechanism, not provider or facility out-of-state location — cross-referenced under RF-13 below rather than counted here, to avoid redefining “location” after seeing the evidence)		
RF-13	Billing for services not rendered or not eligible for reimbursement (narrowed from the original wording)	Directly stated (both anchors + multiple DOJ 2026 cases; also documents billing during beneficiary hospitalization, 2009 OLA — a services-not-rendered/not-eligible variant, cross-referenced from RF-12’s disposition above rather than double-counted as an independent anchor)	Strong — national HHS-OIG program (14 audits, 10+ states including Michigan and Minnesota)	Retain — Strong
RF-14A	Missing or insufficient supporting documentation (split from RF-14)	Directly stated (2009 OLA + 2017 bridge)	Strong — HHS-OIG, full text verified: 18% of a national 10-state Medicaid PCS claims sample had undocumented attendant qualifications	Retain — Strong; assessed independently of RF-14B

RF	Final construct (as revised)	MN derivation anchor status	External convergence outcome	Disposition
			(\$724M inappropriate payments) and 2% had no record services were rendered (\$63M) (OEI-07-08-00430, 2010, p. 9)	
RF-14B	Duplicated, cloned, copied, or fabricated documentation (split from RF-14)	Directly stated (2009 OLA: repeatedly photocopied timesheets with only dates changed; two cases of apparently forged recipient signatures — located during a pre-submission re-review, verified against source text)	Limited (Washington conviction, exact mechanism, press-level source)	Retain — Limited; assessed independently of RF-14A
RF-15	Stolen credentials or unauthorized professional IDs	Not derivable from either anchor (a 2009 OLA passage on an individual holding a second personal care assistant identification number, considered and rejected: it documents duplicate identifier use, not that an identifier was	Limited (same Washington conviction; precise mechanism: misappropriated nursing licenses)	Retain — Limited; the external Washington case already supports the actual construct without stretching the Minnesota passage

RF	Final construct (as revised)	MN derivation anchor status	External convergence outcome	Disposition
		stolen, unauthorized, or fabricated)		
RF-16	Frontline referrals not acted upon or poorly triaged	Directly stated (2017 OLA — clean anchor)	Strong — HHS- OIG, full text verified: 33 of 337 managed care plans (10%) made zero provider-fraud referrals in CY2022, covering 1.6 million enrollees and \$8 billion in payments (OEI-03-22- 00410, 2025, p. 4)	Retain — Strong
RF-17A	Prior enforcement, suspension, termination, exclusion, or sanction history (split from RF-17)	Weak/indirect (enforcement vocabulary only)	Not scored — descriptive/back ground category, not a vulnerability claim	Retain as definitional background; does not inherit RF-17B's rating
RF-17B	Continued participation, billing, or re- entry despite an exclusion, termination, or sanction (split from RF-17)	Directly stated (2009 OLA: a suspended agency's owners re-enrolled under a new agency name; two personal care assistants allowed to continue providing services after DHS removed their	Strong — same two independent national HHS- OIG datasets as RF-06	Retain — Strong; now directly anchored in Minnesota independent of its external convergence

RF	Final construct (as revised)	MN derivation anchor status	External convergence outcome	Disposition
		authorization — located during a pre-submission re-review, verified against source text)		
RF-18	Reactive oversight and insufficient capacity	Directly stated (2017 OLA — the single most solidly anchored indicator in the set)	Limited/related (related national findings; none exclusively on this construct)	Retain — Limited/related
RF-19	Program-design vulnerability arising from low entry barriers, uneven requirements, or oversight capacity that does not keep pace with provider growth (reworded)	Directly stated (2009 OLA — growth paired with a documented management- capacity concern)	None found	Retain — anchor-derived; no external corroboration located
RF-20	Unresolved audit findings or weak corrective action	Directly stated (2009 OLA: “evidence of follow-up by DHS with providers who have been reviewed is virtually nonexistent. We found evidence of follow-up in only 2 of the 26 quality assurance files we reviewed” — located by the second coder,	Strong — GAO, full text verified: 60% of Medicaid single- audit findings (FY2019-2021) were repeat findings, attributed to corrective action not taken, incomplete, or ineffective (GAO-23- 105881, 2023, p. 13)	Retain — Strong

RF	Final construct (as revised)	MN derivation anchor status	External convergence outcome	Disposition
pp. 65-66)				

Second-coder reliability check

Because this derivation audit was otherwise conducted by a single author, an independent second human coder (see Acknowledgments) was given only the three source documents and a blinded list of the 21 active constructs — no access to this Table, this manuscript, or the author’s original classification — and separately classified all 21 rows. Raw agreement was **59.1% (13 of 22 items, including the retired RF-02)**, reported exactly as computed; no chance-corrected statistic is reported, since a 22-item comparison against a single rater does not support a stable kappa estimate. This figure is disclosed as the manuscript’s actual inter-rater reliability, not a higher or more favorable one.

Three of the second coder’s disagreements were not interpretive — they identified passages the original derivation audit’s keyword search had missed entirely. All three were independently verified directly against the source PDF’s extracted text before being accepted, and are reflected above: RF-14B and RF-17B (corrected from unanchored/weak to directly stated) and RF-20 (corrected from reasonable abstraction to directly stated, on a passage documenting that DHS follow-up on quality-assurance reviews was found in only 2 of 26 files examined). Three further disagreements (RF-04, RF-12, RF-15) remain flagged rather than corrected, since each turns on a genuine interpretive judgment about how directly a real, verified passage supports the exact construct wording, not a passage either coder simply missed; the manuscript reports these as open disagreements rather than resolving them by either coder’s authority. Two more of the second coder’s classifications (RF-06, RF-17A) were reviewed and not adopted, because their supporting passages, on inspection, describe a different construct than the one being coded. The full comparison, including every disagreement and the reasoning for each resolution, is in the private development record (`derivation_audit/A7_Human_Coder_Comparison.md`), not part of the public replication package; this appendix reports the outcome, not the underlying working file.

On “candidate for Strong” (fully resolved): four indicators (RF-05, RF-14A, RF-16, RF-20) had, as of an earlier snapshot, identified a national- or multistate-scope federal source with an exact mechanism match, but the underlying document had not yet been read in full and its exact passage pinned — only a title, agency, and headline figure had been confirmed through search. Consistent with this framework’s rule that a convergence rating is never finalized on a search-result summary alone, all four have since been fully verified by direct retrieval and full-text read, with exact page citations, and are now reported as Strong in Table G1 and the main text (Table 1). RF-05 and RF-16 were verified in the prior editorial pass. RF-14A’s original lead — an unlocated “~74% of 2024 Medicaid improper payments attributable to insufficient documentation” figure — could not be traced to a source document and was abandoned; a different, independently verified 2010 HHS-OIG report (OEI-07-08-00430) was instead located and read in full, documenting the same construct in the exact Medicaid Personal Care Services setting (see `derivation_audit/A4_Addendum_02_RF14A_RF20_Verification.md`). RF-20’s GAO source (GAO-23-105881), previously blocked from direct retrieval like several

other GAO sources in this framework (RF-06/RF-09, T-HEHS-96-205, HEHS-97-63, GAO-15-448), was successfully retrieved as a complete PDF via an alternate fetch method and read in full; the other listed GAO sources remain candidates, since re-attempting their retrieval was outside this addendum's scope. No candidate-status indicator remains in this framework. Seven independently verified national sources (RF-06/RF-17B's two-lineage HHS-OIG finding, RF-13's fourteen-audit HHS-OIG rollup, RF-05, RF-16, RF-14A, and RF-20) demonstrate this framework's Strong ratings never depend on unverified candidates.

False-positive safeguards

Every indicator with Limited convergence or better carries a documented benign explanation, an exclusion rule, a minimum-corroboration requirement, and a prohibited interpretation—the same discipline this manuscript already applies to RF-08's shared-contact clusters (Section 3.4; Appendix A). No public-data indicator in this framework licenses an adverse inference on its own.

RF	Benign explanation(s)	Exclusion rule	Minimum corroboration required	Prohibited interpretation
RF-06	Data-entry lag between a provider's exclusion in one state and another state's screening cycle; a common, non-fraudulent surname or entity-name collision	A single missed screening cycle, alone, does not establish evasion	An independent record showing continued billing <i>after</i> the exclusion date, not merely continued licensure	Must not be read as proof the provider knew of the exclusion or acted to evade it
RF-07 / RF-08	Common management company, shared consultant, centralized answering service, franchise arrangement, or a data-entry artifact	A shared <i>commercial</i> registered agent alone is insufficient (a commercial agent may represent hundreds of unrelated entities)	A shared <i>individual</i> registered agent plus a common officer, address, or brand; or same-legal-entity confirmation	Must not be read as evidence of concealed ownership or misconduct absent independent corroboration of common control
RF-13	A documented clerical or systems delay between a	A single isolated claim near a transition date, alone, does not	A pattern of repeated post-ineligibility billing, or an	Must not be read as proof of deliberate fraud absent a

RF	Benign explanation(s)	Exclusion rule	Minimum corroboration required	Prohibited interpretation
	beneficiary's death/discharge and a state's eligibility-file update	establish intent	independent audit/case finding	completed adjudicative finding
RF-14A	Genuine records-retention gaps unrelated to concealment; a documentation requirement that did not exist for a given service type at the time	Absence of a requirement is not equivalent to a violation of one	A documented finding that a requirement existed and was not met	Must not be read as evidence of fabrication (see RF-14B)
RF-14B	Legitimate template use for genuinely similar visits; software-generated formatting that superficially resembles duplication	Formatting similarity alone, without content identity across distinct beneficiaries/dates, is insufficient	An independent audit or case finding of actual duplicated or fabricated content	Must not be read as evidence of missing documentation generally (see RF-14A)
RF-15	Legitimate credential-sharing arrangements within a single licensed entity (e.g., a supervising clinician's number used with authorization)	Use of a professional identifier by an authorized employee of the credential-holder's own entity is not misappropriation	A confirmed, independent record that the credential was used without the holder's authorization	Must not be read as proof of intent absent an adjudicated finding
RF-16 / RF-18	Resource constraints affecting all providers equally, not a targeted failure	A general capacity shortfall, alone, does not establish that a specific credible	A documented instance of a specific referral not acted upon within a stated policy window	Must not be read as evidence of intentional concealment by the oversight agency itself

RF	Benign explanation(s)	Exclusion rule	Minimum corroboration required	Prohibited interpretation
	to act on a specific referral	referral was disregarded		
RF-20	A corrective-action plan genuinely in progress within its stated timeline	A finding repeated the year immediately after it was issued, alone, is not evidence of indifference	A finding still open after its own stated remediation timeline has elapsed	Must not be read as evidence that no corrective action was attempted

Indicators retained with no external corroboration (RF-01, RF-03, RF-10, RF-11, RF-19) and the two recommended for retirement or redefinition (RF-04, RF-12) are not included in the table above because their evidentiary status has not reached the point where a false-positive safeguard is yet operative — RF-04 in particular is recommended for retirement precisely because no available formulation of it has a false-positive safeguard specific enough to prevent an ordinary out-of-state business structure from being flagged as suspicious.

Portability status

This manuscript distinguishes six validity dimensions (Section 2.6): source, content, measurement, population, threshold, and predictive. RF-08 is the only indicator tested across four of the six, and it is where this framework’s central population-validity finding comes from (Section 3.4): the pipeline transferred (technical reproduction), but the threshold required recalibration once facility type was held constant, because Oregon’s mixed four-type population is not equivalent to Washington’s single-type population — a population-validity finding, not merely a threshold-validity one. The new sensitivity analysis (`scripts/build_sensitivity_analysis.py`) extends this finding: Oregon’s threshold-3 signal is carried almost entirely by its Residential Care Facility subtype (5 of 81 RCF facilities, 6.2%), not by Adult Family Homes (0 of 629 at threshold 3) — the same construction rule produces a different substantive result depending on which population it is applied to, independent of any threshold change. The new linkage-robustness analysis (`scripts/build_linkage_robustness.py`) adds a measurement-validity finding for the same indicator, with one correction the verification process itself caught before it could propagate: phone-key normalization is fully robust in Oregon (identical facility counts across four independent normalization variants) and, in Washington, robust once the pre-existing, deliberate exclusion of one non-facility-specific contact number is correctly carried forward from the primary data construction (50 facilities at threshold 3); a naive reconstruction that omits that one exclusion overcounts by exactly three facilities (53) and is reported only as a labeled sensitivity variant, not as evidence that phone-based measurement is unstable. Address-based linkage, by contrast, is genuinely sensitive to normalization choice (Oregon: 59 facilities under strict matching versus 71 under lenient matching at threshold 2, a 20% difference) — meaning RF-08’s phone-based construction is the more measurement-stable of its two possible linkage bases, and the one apparent phone-based discrepancy traces to a single, fully diagnosed data-construction decision rather than to the normalization method itself.

No other indicator in this framework has been operationalized in a form that permits a technical/construct/threshold portability test; RF-13 and RF-06/RF-17B’s Strong external convergence establishes that the underlying *vulnerability* recurs across jurisdictions, which is a different claim from establishing that a specific *measurement* of it would transfer without recalibration.

Validation requirements

Grouped by what would change each remaining disposition, rather than repeated per row: RF-14B and RF-15 require the underlying court record for the one Washington case already identified. RF-07 and RF-09 require a Tier 1 source in the exact Medicaid long-term-care setting, since the sources found so far are either cross-program (Medicare) or below the evidentiary tier this framework treats as sufficient (an academic paper, for RF-07). RF-01, RF-03, RF-10, RF-11, and RF-19 require a qualifying source that has not yet been found despite a completed search and may not exist; their disposition would not change without one. RF-04 and RF-12 require a substantially narrower reformulation before any further search would be productive. No indicator’s predictive validity—whether flagging it actually distinguishes confirmed problem facilities from benign ones—has been tested at any stage of this manuscript; that is Section 4.5’s stated next step for the research program this manuscript begins, not a gap this appendix can close.

Appendix H: Federal/External Corroboration Crosswalk

This appendix reports, source by source, what the external-corroboration search (Appendix G) actually found, tiered by documentary authority rather than presented as uniformly weighted evidence. Three tiers are used throughout: **Tier 1** (audit reports, court filings, plea agreements, judgments, official agency evaluations, final enforcement decisions), **Tier 2** (agency press releases, congressional inquiries, proposed legislation, administrative guidance without facility-level findings), and **Tier 3** (investigative journalism, academic commentary, advocacy reports, non-government database tools). A source’s tier does not change because its finding is favorable: the two Government Accountability Office reports on this same question (T-HEHS-96-205, 1996; HEHS-97-63, 1997) were never elevated past candidate status, because their full text remained blocked from direct retrieval (HTTP 403) despite a favorable search result. RF-06/RF-17B’s Strong rating instead rests on two separate HHS-OIG reports (OEI-06-12-00030, 2015; OEI-03-19-00070, 2020), whose full text was independently downloaded and read before either was counted.

Table H1 reports the verified or candidate sources supporting each indicator with convergence above “none,” their tier, and what would remain to finalize an unverified rating.

RF	Independent source	Tier	Jurisdiction/ scope	Status	Remaining evidence required
RF-05	HHS-OIG, <i>Data on Medicaid Managed Care</i>	1 (verified)	National (largest managed care plan in each	Fully verified , full text read, exact page pinned:	None

RF	Independent source	Tier	Jurisdiction/ scope	Status	Remaining evidence required
	<i>Payments to Providers Are Incomplete and Inaccurate</i> (OEI-02-19-00180, 2021)		of 39 states)	“Most States did not provide complete or accurate payment data in T-MSIS on managed care payments to providers; two States failed to provide any data for January 2020” (p. 1)	
RF-06 / RF-17B	HHS-OIG, <i>Providers Terminated From One State Medicaid Program Continued Participating in Other States</i> (OEI-06-12-00030, 2015)	1 (verified)	National (2,539-provider sample)	Fully verified , full text read, exact pages pinned	None
RF-06 / RF-17B	HHS-OIG, <i>States Could Do More To Prevent Terminated Providers From Serving Medicaid Beneficiaries</i> (OEI-03-19-00070, 2020)	1 (verified)	National (9,047-provider CMS dataset, independently collected)	Fully verified , full text read	None
RF-06 / RF-17B	GAO, <i>Fraud and Abuse:</i>	1 (candidate)	National	Not required for the Strong	Full-text access

RF	Independent source	Tier	Jurisdiction/ scope	Status	Remaining evidence required
	<i>Providers Excluded From Medicaid Continue to Participate</i> (T-HEHS-96-205, 1996) and <i>Medicaid Fraud and Abuse: Stronger Action Needed</i> (HEHS-97-63, 1997)			rating (already met by the two HHS-OIG sources above); retained for historical depth only, pending full-text access (blocked, HTTP 403 on direct retrieval)	
RF-07	HHS-OIG, <i>Some Selected Skilled Nursing Facilities Did Not Comply With Medicare Requirements for Reporting Related-Party Costs</i> (2024)	1 (candidate)	National, Medicare SNF setting	Confirmed via search; cross-program setting caveat applies	Full-text verification; a Medicaid-HCBS-setting equivalent
RF-07	Gandhi & Olenski, <i>Tunneling and Hidden Profits in Health Care</i> (NBER working paper)	3 (confirmed)	Multistate, incl. a Michigan-specific finding	Academic working paper, not (yet) peer-reviewed — supports mechanism plausibility and literature discussion only; cannot independently elevate	Not applicable at this tier; a government source citing the same underlying cost-report data could be scored separately

RF	Independent source	Tier	Jurisdiction/ scope	Status	Remaining evidence required
				convergence per the frozen protocol's tier rule	
RF-09	GAO, <i>Medicare Program: Additional Actions Needed to Improve Eligibility Verification of Providers and Suppliers</i> (GAO-15-448, 2015)	1 (candidate)	National, Medicare/PE COS setting	Exact mechanism match (a Commercial Mail Receiving Agency address verified as an authentic practice location); blocked from full-text retrieval	An accessible official copy; a Medicaid-HCBS-setting equivalent
RF-13	HHS-OIG, <i>Multiple States Made Medicaid Capitation Payments to Managed Care Organizations After Enrollees' Deaths</i> (A-04-21-09005, 2023), rolling up 14 prior state audits, 2009–2019	1 (verified)	National, 10+ states including Michigan and Minnesota	Fully verified via direct retrieval	None
RF-14A	HHS-OIG, <i>Inappropriate Claims for Medicaid Personal</i>	1 (verified)	National (10-state sample of Medicaid PCS claims)	Fully verified , full text read, exact page pinned:	None

RF	Independent source	Tier	Jurisdiction/ scope	Status	Remaining evidence required
	<i>Care Services</i> (OEI-07-08-00430, 2010)			“Attendant qualifications were undocumented for 18 percent of Medicaid PCS claims, resulting in \$724 million in inappropriate payments” and 2% of claims had no record services were rendered (\$63M) (p. 9)	
RF-14B / RF-15	Washington Attorney General’s Office, Medicaid Fraud Control Division, conviction announcement (jury verdict, 2026-05-28, all counts)	2	Washington (in-sample study state)	Verified as an official press release reporting a completed conviction; underlying court record not yet obtained	Underlying plea/judgment/sentencing record
RF-16	HHS-OIG, <i>Some Medicaid Managed Care Plans Made Few or No Referrals of Potential Provider Fraud</i>	1 (verified)	National (337 plans, 13 states)	Fully verified , full text read, exact page pinned: “33 of 337 plans (10 percent) reported that they did not make any	None

RF	Independent source	Tier	Jurisdiction/ scope	Status	Remaining evidence required
	(OEI-03-22-00410, 2025)			provider referrals to any entity in CY 2022. Combined, these 33 plans covered 1.6 million enrollees and received \$8 billion in payments from 13 States” (p. 4)	
RF-20	GAO, <i>Medicaid Program Integrity: Opportunities Exist for CMS to Strengthen Use of State Auditor Findings and Collaboration</i> (GAO-23-105881, 2023)	1 (verified)	National (50 states + D.C.; 923 single-audit findings, FY2019-2021)	Fully verified , full text read, exact page pinned: “60 percent of Medicaid single audit findings... were repeat findings... [because] a corrective action was not taken, was not completed, or was ineffective” (p. 13)	None

Table H2 reports four sources identified in an earlier research pass and formally tiered here for the first time, mapped to the specific indicators they actually bear on rather than presented as general contextual reading.

RF	Source	Tier	What it establishes
RF-05, RF-18	GAO-18-179 (2018): 26 of 48 state Medicaid agencies	1	An independent, pre-2026 national finding of the same

RF	Source	Tier	What it establishes
	could not report critical-incident counts in assisted living		administrative-data-incompleteness and reactive-oversight pattern this framework documents in Washington and Oregon separately
RF-05, RF-09	GAO-26-107884 (2026): assisted living is not a uniformly defined provider type and cannot be consistently identified across Medicare/Medicaid claims	1	Independent, national confirmation that facility-type definitional inconsistency—the same phenomenon underlying this manuscript’s own Oregon population-validity finding (Section 3.4; Appendix G’s portability discussion)—is not specific to Washington and Oregon
RF-07, RF-08	ProPublica, <i>Find Out Who Owns a Nursing Home</i> (Talbot, 2023) and its 2026 update documenting the Benjamin Landa ownership-discrepancy example	3	A downstream-use precedent for public ownership-linkage tools (Section 4.1’s discussion of who can use the framework) and a documented case where affiliated-entity disclosure understated true common ownership—consistent with this framework’s own finding that a shared-commercial-registered-agent signal alone is insufficient

RF	Source	Tier	What it establishes
RF-03, RF-08	Minnesota HF4144/SF4279 (94th Legislature, 2025–2026, pending in committee): would bar new small assisted-living facilities within 650 feet of an existing same-type facility	2	corroboration Legislative attention to facility concentration as a policy concern, offered as external- validity context for RF-03/RF-08's concentration indicators (Section 4.4)—not a validation of either indicator, since proposed legislation is not an evidentiary finding

Per the tier-ceiling rule established in this framework's search protocol, the ProPublica and Minnesota-legislation entries in Table H2 support contextual discussion in Sections 4.2 and 4.4 only; neither elevates RF-03, RF-07, or RF-08 beyond the convergence rating each already carries on the strength of the Tier 1 and Tier 2 sources in Table H1.

Oregon's shared-contact corroboration (RF-08)

Unlike the sources in Tables H1–H2, RF-08 was tested directly rather than searched for in the secondary literature: all 49 of Oregon's shared-telephone-number facility clusters were individually checked against Oregon's public business registry. Thirty-six clusters (73%) reflected an affiliation already visible in the facility names themselves (e.g., a shared assisted-living brand operating adjoining assisted-living and memory-care licenses); eleven additional clusters (22%)—not visible from the facility names at all—were corroborated through registry evidence: eight through a common officer, owner, or manager, two through a common residential or business address (kept analytically distinct, since a shared address alone does not establish common control), and one through outright common legal-entity identity. One cluster (2%) yielded no corroborating public connection, and one (2%) remained indeterminate because the operator's surname was too common to confidently resolve identity in the registry. The eleven non-obvious corroborations are the result that matters for RF-08's standing as an entity-resolution proxy: they demonstrate the shared-contact signal surfacing real relationships the public-facing facility data does not itself disclose, while the one non-corroborated and one indeterminate cluster demonstrate why the proxy requires case-level corroboration rather than standing on its own as a review-priority finding. Full cluster-level detail, evidence-class coding, and the registry citations underlying each corroborated relationship are in the released anonymized data (`anonymized_data/oregon/A5_OR_Cluster_Corroboration_ANON.csv`); the raw facility names and registry records reviewed to produce that classification are not part of the public replication package, consistent with this manuscript's general treatment of privacy-sensitive entity-resolution steps (Section 2.6).

Supplementary References

Sources cited in Appendices G and H that do not appear in the main manuscript's reference list. Full source-by-source tiering and verification status for each is reported in Appendix H (Tables H1–H2); access dates for all entries below are recorded in SOURCE_MANIFEST.md.

Gandhi, A., & Olenski, A. (n.d.). *Tunneling and hidden profits in health care* [Working paper]. National Bureau of Economic Research. <https://www.nber.org/papers/>

U.S. Government Accountability Office. (1996, June 12). *Fraud and abuse: Providers excluded from Medicaid continue to participate* (T-HEHS-96-205) [Testimony]. <https://www.gao.gov/products/t-hehs-96-205>

U.S. Government Accountability Office. (1997, March 3). *Medicaid fraud and abuse: Stronger action needed to remove excluded providers from federal health programs* (HEHS-97-63). <https://www.gao.gov/products/hehs-97-63>

U.S. Government Accountability Office. (2015, June 25). *Medicare program integrity: Increasing consistency of contractor requirements may improve administrative efficiency* (GAO-15-448). <https://www.gao.gov/products/gao-15-448>

U.S. Government Accountability Office. (2018, March). *Medicaid assisted living services: Improved federal oversight of beneficiary health and welfare is needed* (GAO-18-179). <https://www.gao.gov/products/gao-18-179>

U.S. Department of Health and Human Services, Office of Inspector General. (2010, December). *Inappropriate claims for Medicaid personal care services* (OEI-07-08-00430). <https://oig.hhs.gov/oei/reports/oei-07-08-00430.pdf>

U.S. Government Accountability Office. (2023, September 21). *Medicaid program integrity: Opportunities exist for CMS to strengthen use of state auditor findings and collaboration* (GAO-23-105881). <https://www.gao.gov/assets/gao-23-105881.pdf>

U.S. Government Accountability Office. (2026). *Medicaid assisted living: Inconsistent provider-type definitions limit oversight* (GAO-26-107884). <https://www.gao.gov/products/gao-26-107884>

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U.S. Department of Health and Human Services, Office of Inspector General. (2021, March). *Data on Medicaid managed care payments to providers are incomplete and inaccurate* (OEI-02-19-00180). <https://oig.hhs.gov/oei/reports/OEI-02-19-00180.pdf>

U.S. Department of Health and Human Services, Office of Inspector General. (2020). *States could do more to prevent terminated providers from serving Medicaid beneficiaries* (OEI-03-19-00070). <https://oig.hhs.gov/oei/reports/oei-03-19-00070.asp>

U.S. Department of Health and Human Services, Office of Inspector General. (2023). *Multiple states made Medicaid capitation payments to managed care organizations after enrollees' deaths* (A-04-21-09005). <https://oig.hhs.gov/reports/all/2023/>

U.S. Department of Health and Human Services, Office of Inspector General. (2024). *Some selected skilled nursing facilities did not comply with Medicare requirements for reporting related-party costs*. <https://oig.hhs.gov/reports/all/2024/>

U.S. Department of Health and Human Services, Office of Inspector General. (2025, August 27). *Some Medicaid managed care plans made few or no referrals of potential provider fraud* (OEI-03-22-00410). <https://oig.hhs.gov/documents/evaluation/10912/OEI-03-22-00410.pdf>

Washington State Attorney General's Office, Medicaid Fraud Control Division. (2026, May 28). *AG's office obtains conviction on all counts in Medicaid fraud trial* [Press release]. <https://www.atg.wa.gov/news/news-releases/ag-s-office-obtains-conviction-all-counts-medicaid-fraud-trial>